DAEKYUNG BEND CO.,LTD.

INSPECTION CERTIFICATE

(EN10204 3.1)

115, Eosil-ro, Yangsan-Si,
Gyeongsangnam-Do,
Korea
TEL : 82-055-364-1800
FAX : 82-055-362-7074

Customer	월드원플랜에탈	**Certificate. No.**	YR-26-07015-6
P.O. No.	ES260709-1974	**Date of Issue**	2026.07.09
Hull No.(Project No.)	N/A	**Dimension & VI**	GOOD
Spec. for Material	ASTM A312 TP304	**Spec. for Inspection**	KS B1543 STS304-W /B1500

1. ITEM DESCRIPTION

DKB LOT No	Item No	Heat No	Description	Q'TY(P'cs)	Starting Material	NDE(Report No.)
KGA0589	31	SE51511	T(R)-S10S-50A X 15A-STS316L-W	1	WELDED PIPE	N/A
KGA0589	32	SE51511	90E(L)-S10S-25A-STS316L-W	12	WELDED PIPE	N/A
KGA0589	33	SE51511	90E(L)-S10S-40A-STS316L-W	12	WELDED PIPE	N/A
KGA0589	34	SE51511	90E(L)-S10S-50A-STS316L-W	6	WELDED PIPE	N/A

* Heat Treatment Condition : 1050℃ Solution Annealing & Water Quenching

2. CHEMICAL COMPOSITION(%)

Heat No.	Spec	C	Si	Mn	P	S	Ni	Cr							
	Max	0.080	1.00	2.00	0.045	0.030	11.00	20.00							
	Min						8.00	18.00							
SE51511	L	0.023	0.42	1.47	0.032	0.002	8.09	18.12							
SE51511	L	0.023	0.42	1.47	0.032	0.002	8.09	18.12							
SE51511	L	0.023	0.42	1.47	0.032	0.002	8.09	18.12							
SE51511	L	0.023	0.42	1.47	0.032	0.002	8.09	18.12							

* L : Ladle Analysis

3. MECHANICAL PROPERTIES

	TENSILE TEST					HARDNESS TEST	CHARPY-V IMPACT TEST (Test Temp : ℃)									
Heat No	Unit	Y.S(0.2%)	T.S	E.L	R.A	TEST METHOD: HB	Specimen Size	Direction Position	ABSORBED ENERGY (Unit : J)				LATERAL EXPANSION (Unit : mm)			
		MPa	MPa	%	%				Min: (S)		(AVR)		Min : mm			
	Max	-	-	-	-	-										
	Min	205	515	35	-	-			1	2	3	AVR	1	2	3	AVR
SE51511	L,B	288	654	55												
SE51511	L,B	288	654	55												
SE51511	L,B	288	654	55												
SE51511	L,B	288	654	55												

* L : Longitudinal, T : Transvers, B : Base Metal, W : Weld Metal

Heat No	PMI	FLATTENING	CORROSION TEST	HYDROSTATIC TEST	FERRITE	GRAIN SIZE
SE51511	GOOD	N/A	N/A	N/A	N/A	N/A
SE51511	GOOD	N/A	N/A	N/A	N/A	N/A
SE51511	GOOD	N/A	N/A	N/A	N/A	N/A
SE51511	GOOD	N/A	N/A	N/A	N/A	N/A

REMARK

We certify this material has been manufactured and examined in accordance with all requirements of the specification and the results of all examination are acceptable

S. J. KIM

HEAD OF QUALITY MANAGEMENT